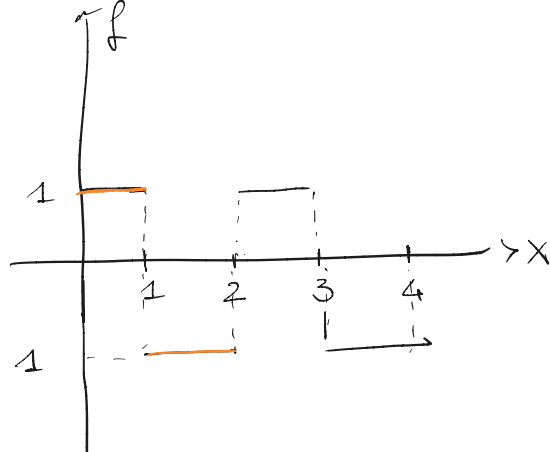




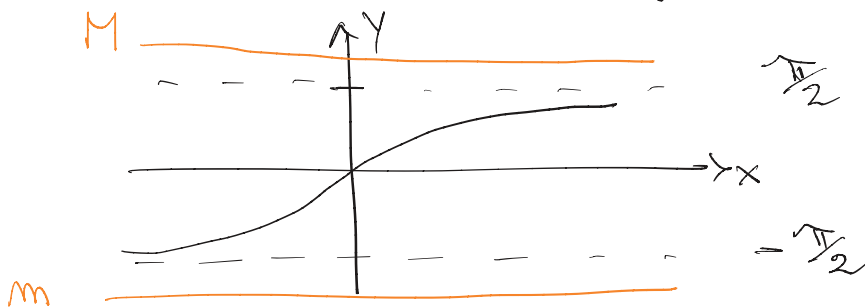
è periodo di f
 $\bar{e} \quad T = 2$

DEF Una funzione $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$
 è limitata superiormente se
 $f(A) \subseteq \mathbb{R}$ è (inferiormente)
 limitato superiormente
 (inferiormente)

Esempio: $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = \arctg x$

$$f(\mathbb{R}) = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

f è una funzione limitata
 superiormente e inferiormente



$$\exists M \in \mathbb{R} \text{ t.c. } f(x) \leq M \quad \forall x \in A$$

$$f: A \subseteq \mathbb{R} \rightarrow \mathbb{R}$$

$$\forall M \in \mathbb{R} \\ \exists x \in A \text{ t.c. } \\ f(x) > M$$

limitata superiormente
limitata inferiormente

f è limitata se è limitata sia superiormente che inferiormente

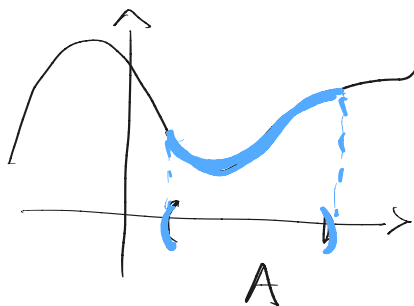
DEF

$$\sup_A f := \sup f(A)$$

$$\inf_A f := \inf f(A)$$

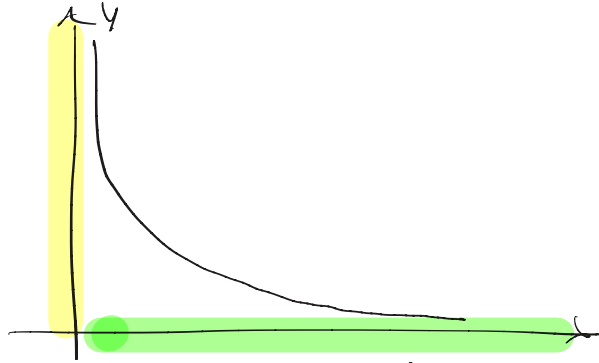
$$\max_A f := \max f(A)$$

$$\min_A f := \min f(A)$$



Esempio: $f: A \subseteq \mathbb{R} \rightarrow \mathbb{R} \quad A = (0, +\infty)$
 $f(x) = \frac{1}{x}$

trovare sup e inf
di f in A



$$\sup_A f = \sup_{(0, +\infty)} f = \sup_{x \in (0, +\infty)} f(x) = \sup_{x \in (0, +\infty)} f((0, +\infty)) = +\infty$$

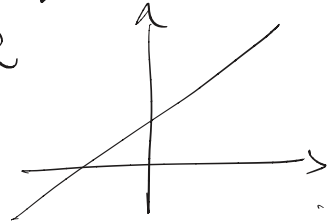
$$\inf_A f = \inf_{x \in (0, +\infty)} f(x) = \inf_{x \in (0, +\infty)} f((0, +\infty)) = 0$$

f non è limitata superiormente
e limitata inferiormente
non ammette max e minimo

Funzione Elementari $f: \mathbb{R} \rightarrow \mathbb{R}$

(a) funzioni lineari (/affini)

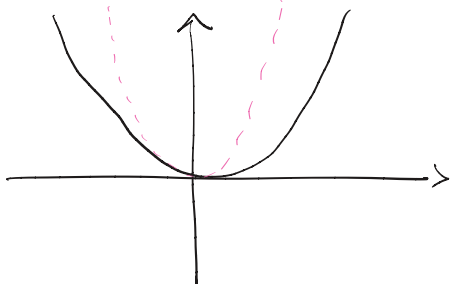
$$f(x) = ax + b \quad a, b \in \mathbb{R}$$



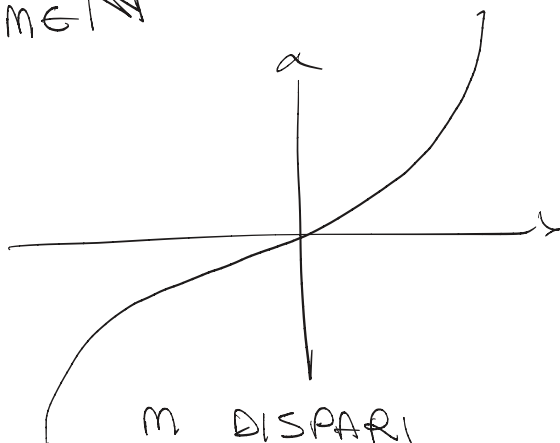
(b) Potenze

$$* f(x) = x^m$$

$$m \in \mathbb{N}$$



m PARI



m DISPARI

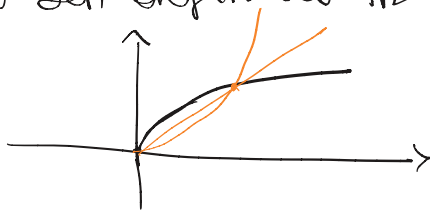
$$* f(x) = x^{p/q}$$

$p, q \in \mathbb{N}$ coprimi, $q \neq 0$

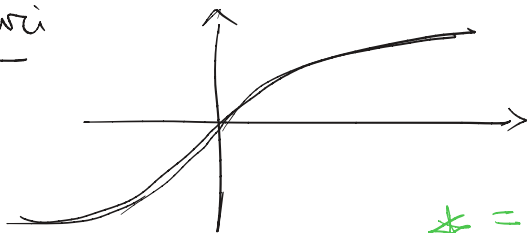
se q è pari, f è ben definita in $[0, +\infty)$

se q è dispari, f è ben definita in $\mathbb{R}$

$p=1$ $f(x) = x^{1/q}$
 q è PARI



$q=3$ dispari
 $f(x) = \sqrt[3]{x}$



$$* = \frac{1}{x^{p/q}}$$

$$* f(x) = x^{-p/q}$$

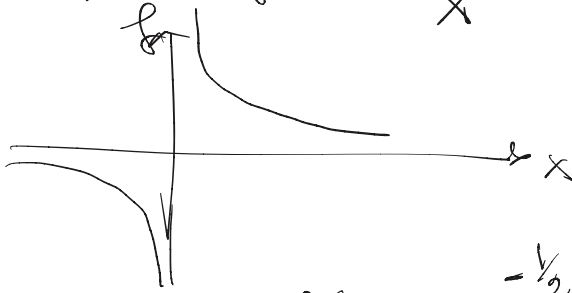
$p, q \in \mathbb{N}$, $q \neq 0$

se q è pari, f è ben definita in $(0, +\infty)$

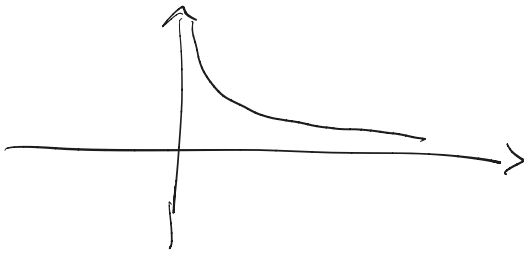
se q è dispari, f è ben definita in $\mathbb{R} \setminus \{0\}$

$\mathbb{R} \setminus \{0\}$

esempio $f(x) = \frac{1}{x}$ è ben definita in $\mathbb{R} \setminus \{0\}$



$$f(x) = x^{-1/2} = \frac{1}{\sqrt{x}} \quad (0, +\infty)$$

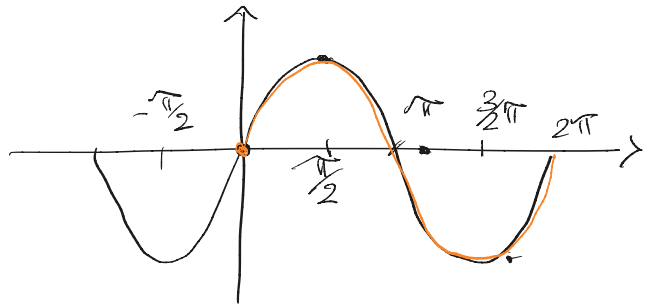


(c) Goniometriche (o circolari)

$$f(x) = \sin x$$

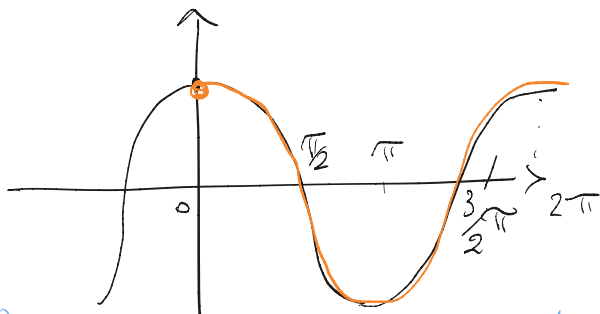
- periodica di periodo 2π
- è limitata

$$f(\mathbb{R}) = [-1, 1]$$

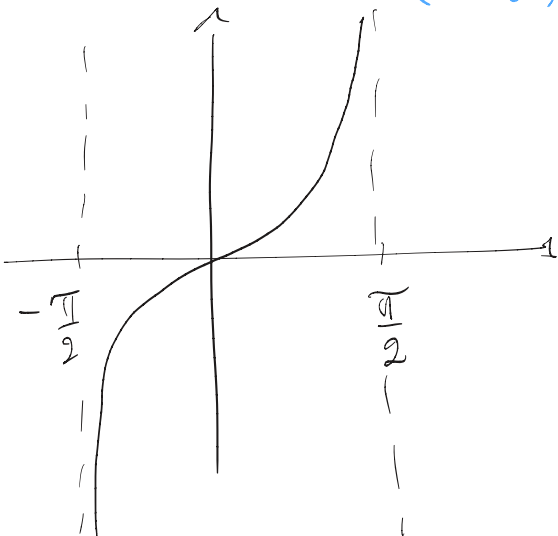


$$\sup_{\mathbb{R}} f = \max_{\mathbb{R}} f = 1 \quad ; \quad \inf_{\mathbb{R}} f = \min_{\mathbb{R}} f = -1$$

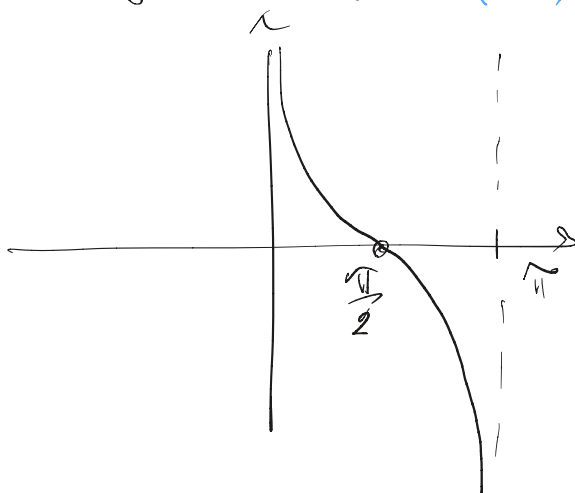
$$f(x) = \cos x$$



$$f(x) = \tan x \left(= \frac{\sin x}{\cos x} \right)$$



$$f(x) = \cot x = \left(\frac{\cos x}{\sin x} \right)$$



è ben definita in

$$\mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi \mid k \in \mathbb{Z} \right\} = A$$

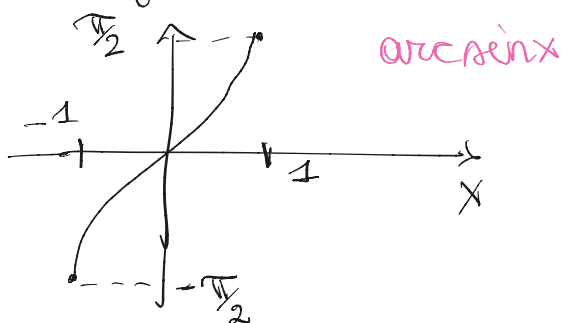
è ben definita in

$$\mathbb{R} \setminus \{ k\pi \mid k \in \mathbb{Z} \}$$

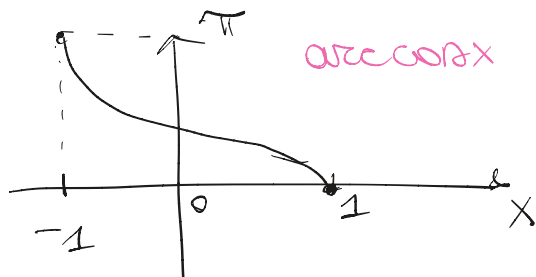
- funzioni periodiche di periodo π ($T = \pi$)
- funzioni illimitate (sia sup. che inferiori.)

$$\sup_A f = +\infty \quad \inf_A f = -\infty$$

(d) funzioni circolari inverse

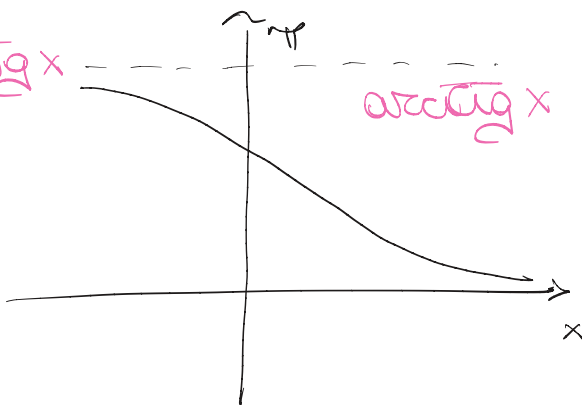
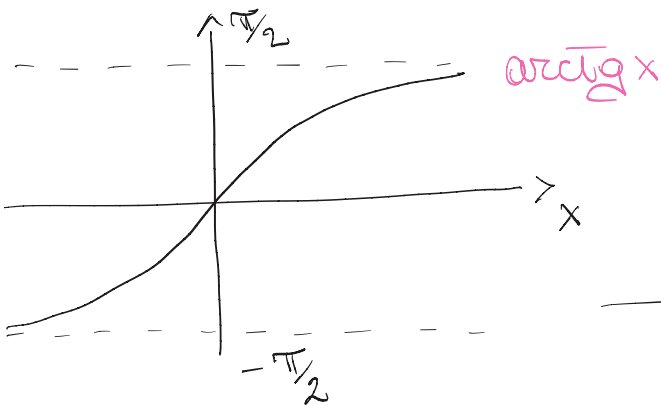


$$f: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$



$$f: [-1, 1] \rightarrow [0, \pi]$$

per ottenere queste funzioni ho ristretto
il $\sin x$ a $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ e il $\cos x$ a $[0, \pi]$ in
modo che fossero iniettive e invertibili



per ottenere queste funzioni ho ristretto
il $\lg x$ a $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ e il $\cotg x$ a $[0, \pi]$ in
modo che fossero iniettive e invertibili

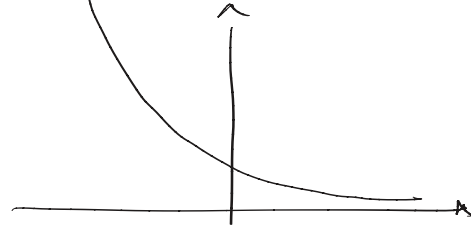
(e) L' esponenziale e il logaritmo

$$f: \mathbb{R} \rightarrow (0, +\infty) \quad f(x) = a^x$$

$$a > 1$$



$$0 < a < 1$$



limitata inferiormente $\inf_{\mathbb{R}} f = 0$

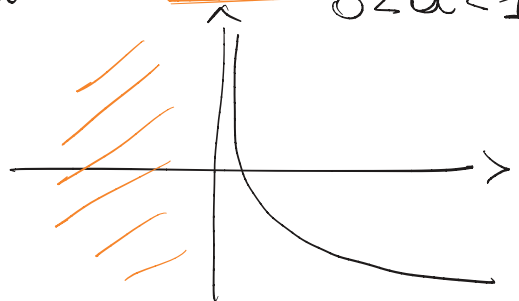
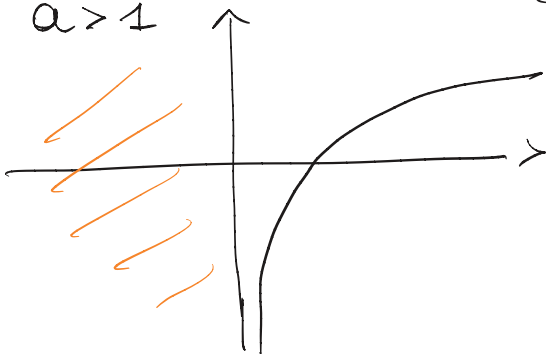
illimitata superiormente $\sup_{\mathbb{R}} f = +\infty$

con questa scelta di codominio, la funzione è biettiva

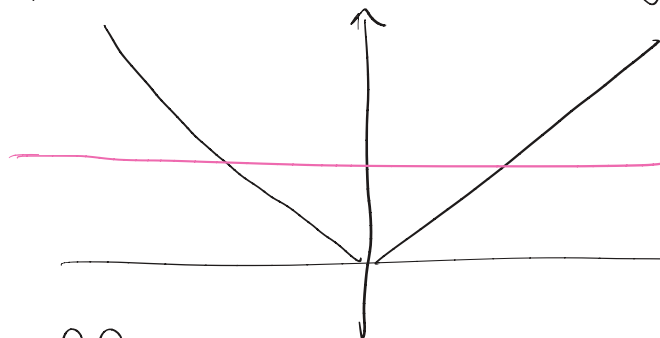
$f^{-1}: (0, +\infty) \rightarrow \mathbb{R}$ per definizione

$$f^{-1}(y) = x \quad a^x = y \rightarrow x = \log_a y$$

$$\sim \text{ } \quad a > 1 \quad f(x) = \log_a x \quad f: (0, +\infty) \rightarrow \mathbb{R} \quad 0 < a < 1$$



(g) Valore assoluto ($\oplus$ modulo)



$$f(x) = |x|$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

pari

$$\inf_{\mathbb{R}} f = \min_{\mathbb{R}} f = 0$$

$$\sup_{\mathbb{R}} f = +\infty$$

$$f(\mathbb{R}) = [0, +\infty)$$

← limitare inf.

← lim. superior.

(h) La parte intera

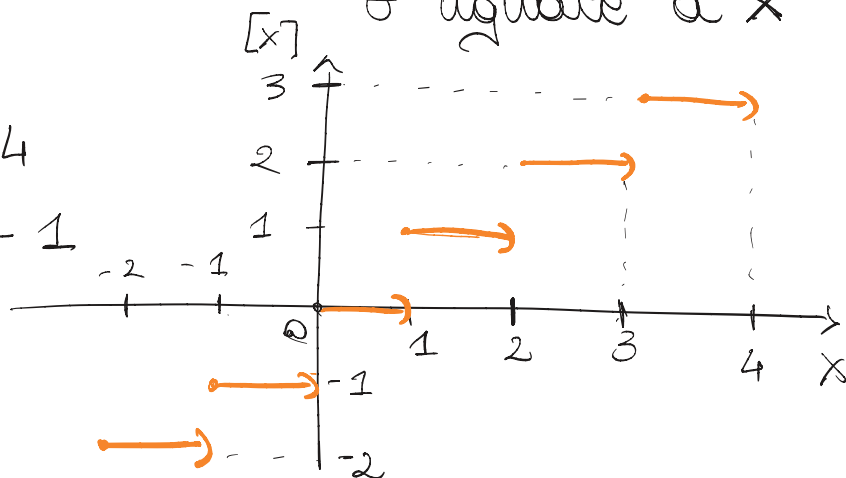
Dato $x \in \mathbb{R}$, $[x] :=$ il più grande
intero più piccolo
 $\oplus$ uguale a x

$$[\pi] = 3$$

$$[2] = 2$$

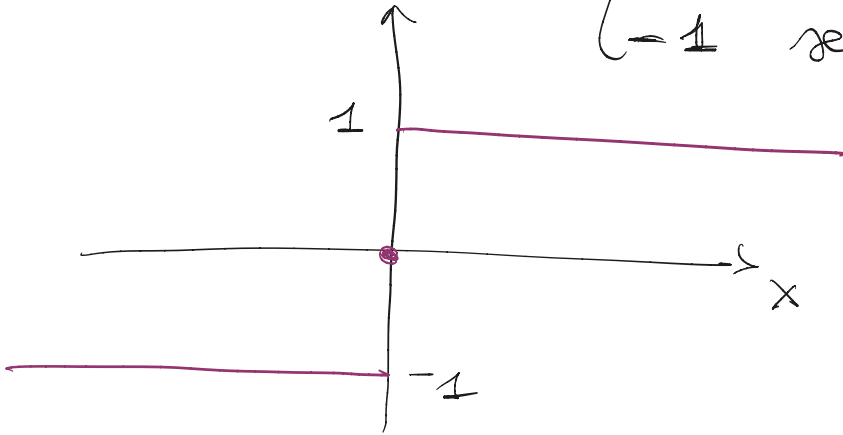
$$[-\pi] = -4$$

$$[-0,5] = -1$$



(i) Funzione segno

$$f(x) = \operatorname{sgn}(x) := \begin{cases} 1 & \text{se } x > 0 \\ 0 & \text{se } x = 0 \\ -1 & \text{se } x < 0 \end{cases}$$



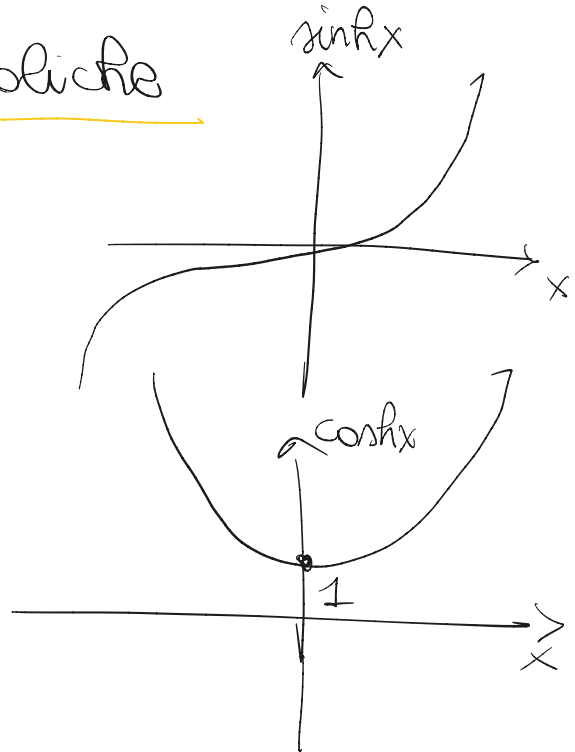
(l) Funzioni iperboliche

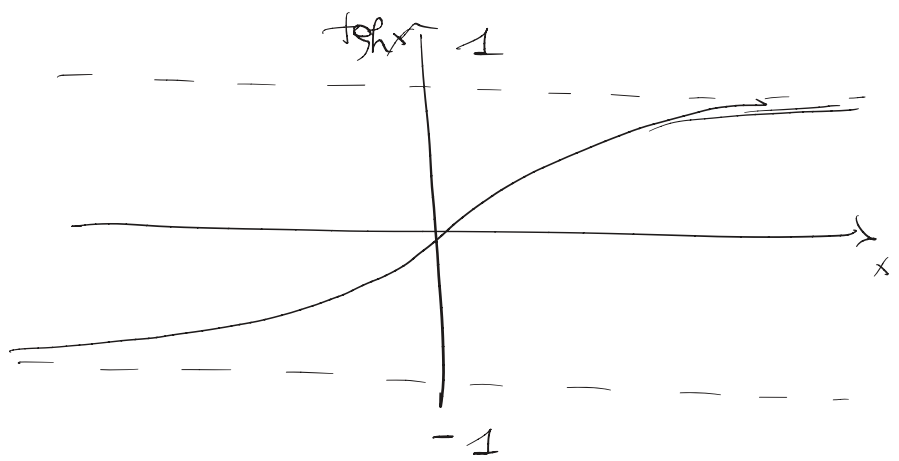
$$\sinh x := \frac{e^x - e^{-x}}{2}$$

$$\cosh x := \frac{e^x + e^{-x}}{2}$$

$$\tanh x := \frac{\sinh x}{\cosh x}$$

e = numero di Nepero





$x^2 - y^2 = 1 \leadsto$ iperbole equilatera

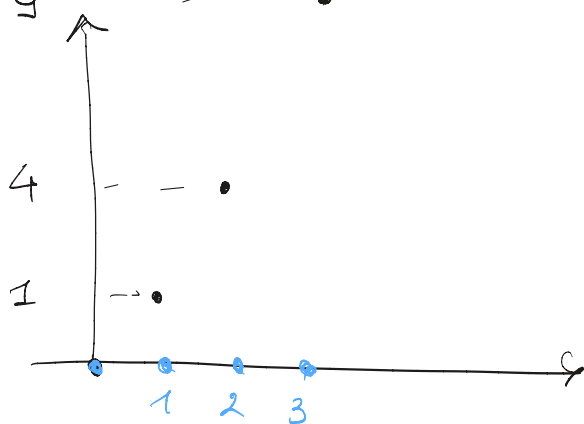
$$(\cosh x)^2 - (\sinh x)^2 = 1$$

SUCCESSIONI!

DEF: Una successione è una funzione
 $f: \underline{\mathbb{N}} \longrightarrow \mathbb{R}$

Notazione: le indichiamo con $a_m (= f(m))$

Esempio: $a_m = m^2$ $m \in \mathbb{N}$ pedice



$$a_0 = 0$$

$$a_1 = 1$$

$$a_2 = 4$$

$$a_3 = 9$$

- a_m sono gli elementi della successione
- la successione la indico come insieme dei suoi elementi: $\{a_m\}_{m \in \mathbb{N}}$
- $\text{Im } a_m = \{ \text{valori della successione} \}$

Es $a_m = (-1)^m \quad m \in \mathbb{N}$

$$\{a_m\}_{m \in \mathbb{N}} = \{a_0, a_1, a_2, \dots\}$$

$$\text{Im } a_m = \{-1, 1\}$$

$$a_0 = 1$$

$$a_1 = -1$$

$$a_2 = 1$$

$$* n! = a_n \quad n! := n(n-1)(n-2)\dots \cdot 1$$

$$0! = 1$$

$$a_0 = 1; a_1 = 1; a_2 = 2; a_3 = 6; a_4 = 24$$

DEF: $\{a_m\}$ è limitata superiormente
(inferiormente)

$\& \text{Im } a_m \subseteq \mathbb{R}$ è lim. superiorum.
(inferiorum.)

$$* \exists M \in \mathbb{R} : a_m \leq M \quad \forall m \in \mathbb{N}$$

$$* \exists m \in \mathbb{R} : a_m \geq m \quad \forall m \in \mathbb{N}$$

Esercizio: $a_m = m^3$ $a_0 = 0$

$$a_1 = 1$$

$$a_2 = 8$$

• a_m è limitata
inferiormente

• a_m è limitata superiormente

$$\forall M \in \mathbb{R} : \exists m_0 \in \mathbb{N} \text{ t.c. } a_{m_0} > M$$

verifichiamo: fisso $M \in \mathbb{R}$ arbitrario

$$m^3 > M \iff m > \sqrt[3]{M}$$

per esempio prendo $m_0 = \lceil \sqrt[3]{M} \rceil + 1$